CIRCUIT DEBUGGING ASSESSMENT

ECA0405 – ANALOG CIRCUITS | CO3 | ASSESSMENT – 2

Student Name	S. Visnu	**Register No.**	192512423
Class / Section	Slot C	**Date**	15/08/2026
Faculty Name	Dr. Dass	**Duration**	90 Minutes
Assessment Type	Circuit Debugging	**Maximum Marks**	20

Assessment Objective:

To identify, analyze, and correct errors in an analog circuit, justify the debugging approach, verify the corrected circuit operation, and relate the result to CO3

Question:

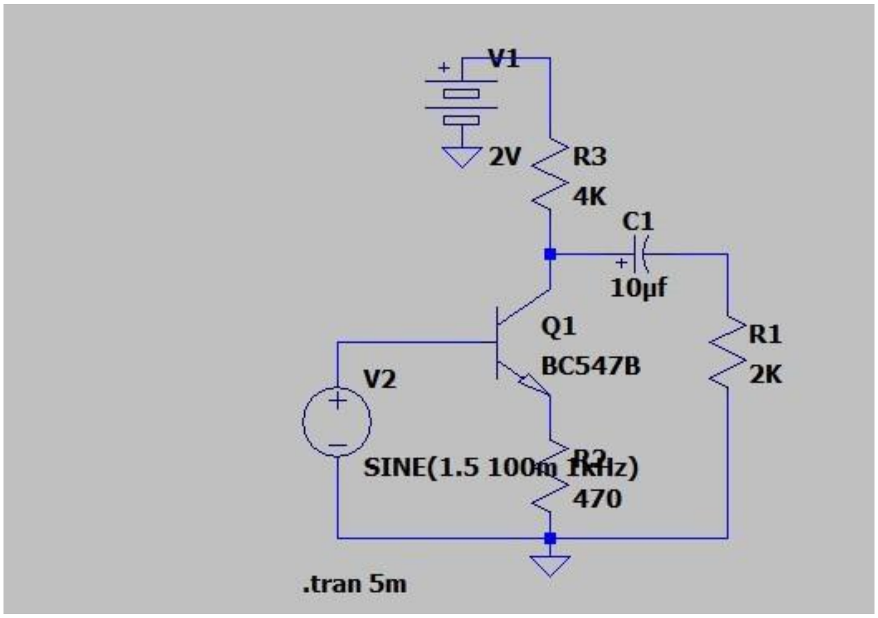
CO3 : Examine the frequency response characteristics of BJT and FET amplifiers at low and high frequencies to determine bandwidth, gain variation, and overall amplifier performance

CO3 AT2 Circuit Debugging:

Description:

- Each student will be assigned one Circuit debugging problem 20 marks inside class. Analyze and debug one faulty analog circuit.
- Simulate the corrected circuit using LTspice and verify results.
- Document the debugging process, observations, and conclusions.
- Upload LTspice files, simulation results, and report to GitHub.
- Evaluation is based on fault identification, debugging approach, simulation accuracy, documentation, and GitHub submission.

Circuit Debugging Record

<table>
<tr><th>Item</th><th>Student Entry</th></tr>
<tr><td>1. Given Circuit / Circuit Diagram</td><td>
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<tr><td>2. Expected Circuit Operation</td><td>The diagram depicts a Bipolar Junction Transistor (BJT) configured as a Common Emitter (CE) amplifier (note: problem text states Common Collector, but schematic physical connections define it as CE). The expected operation is linear signal amplification. This requires the BJT to be biased centrally in the active region to provide sufficient voltage headroom ($\square_{CE}$) for the output AC signal to swing symmetrically without clipping</td></tr>
<tr><td>3. Observed Fault / Symptom</td><td>The supply voltage is accidentally reduced to 2V. Under transient simulation with a 100mV peak, 1kHz input signal (DC offset 1.5V), the amplifier exhibits a severe loss of gain and the output waveform is heavily distorted/flattened, failing to accurately reproduce the input signal.</td></tr>
<tr><td>4. Suspected Component / Connection</td><td>The primary DC supply voltage node ($\square_{cc}$).</td></tr>
<tr><td>5. Debugging Analysis / Calculation</td><td>For linear amplification, the BJT must operate in the active region (Base-Emitter forward-biased, Base-Collector reverse-biased). With $\square_{cc}$ severely reduced to 2V, the quiescent operating point (Q-point) shifts drastically. The standard voltage drop across the collector resistor ($\square_C = 4$ kΩ) forces the DC collector voltage ($\square_C$) dangerously close to ground. When $\square_C$ drops below the required base voltage ($\square_B$), the Base-</td></tr>
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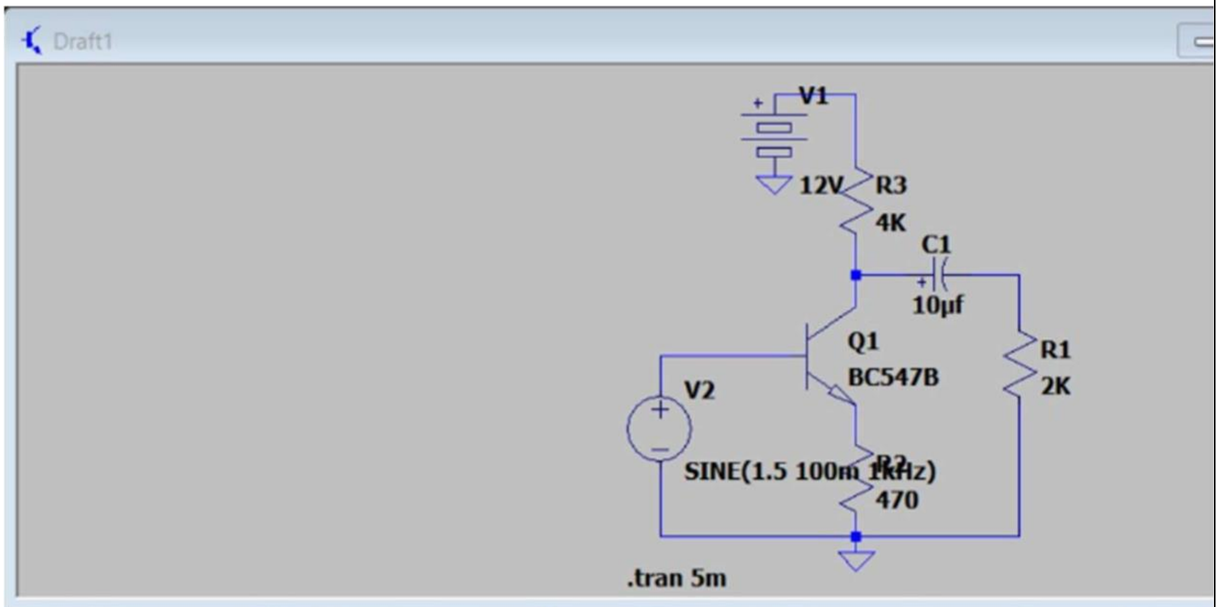
<tr><td></td><td>Collector junction becomes forward-biased. This drives the transistor into the saturation region ($\square_{CE(sat)} \approx 0.2$ V), entirely eliminating the available voltage headroom necessary for signal swing.</td></tr>
<tr><td>6. Error Identified</td><td>Insufficient DC supply voltage ($\square_{cc} = 2$ V). This starves the circuit, preventing proper biasing and forcing the transistor out of the active linear region and into saturation, causing severe output distortion.</td></tr>
<tr><td>7. Correction Made</td><td>Increased the DC supply voltage ($\square_{cc}$) from the faulty 2V to a standard operational value of 12V. This restores the Q-point closer to the center of the AC load line.</td></tr>
<tr><td>8. Corrected Circuit Diagram</td><td>


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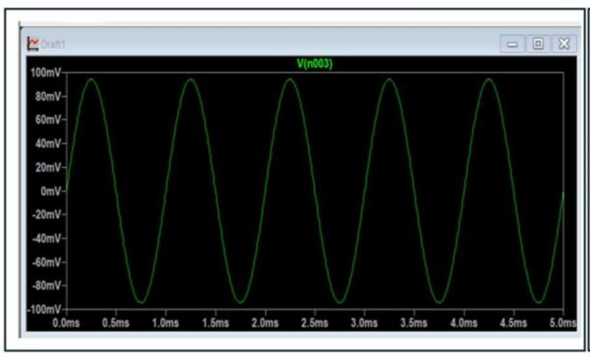
<tr><td>9. Verification / Simulation Output</td><td>
<table>
<tr><td>Faulty output given in the question</td><td>Corrected output</td></tr>
<tr><td>


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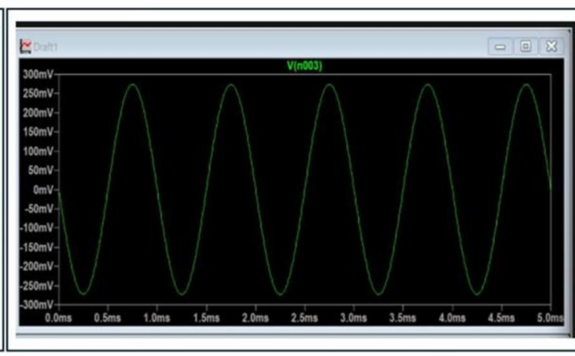

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Observation: At 12V, the output waveform V(n003) successfully amplifies the 100mV peak input signal to approximately 270mV peak, yielding a clean, unclipped sine wave with a voltage gain of ~2.7.
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10. Final Result & Technical Justification	By restoring V_{cc} to a nominal 12V, the transistor's operating point is shifted back into the active region. This provides sufficient V_{CE} headroom for the output signal to swing freely during both positive and negative half-cycles. The overall amplifier performance is restored, demonstrating expected gain and eliminating the severe distortion caused by premature saturation.

Marks Distribution

Assessment Criterion	Marks	Marks Awarded	Faculty Remarks
Problem identification & circuit understanding	4		
Fault analysis & technical calculations	5		
Correct identification of error	3		
Effective circuit correction	3		
Verification / output validation	3		
Technical explanation & documentation	2		
Total	20		